

CS 352

Computer Graphics

Lecture 6

Design of Ellipse Algorithms

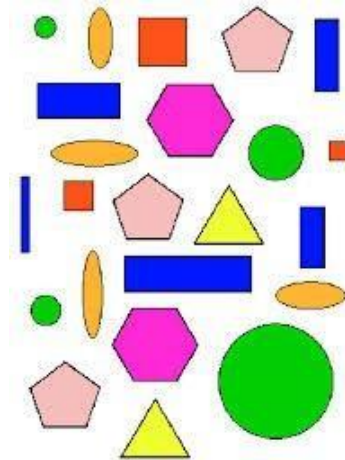


Output primitives

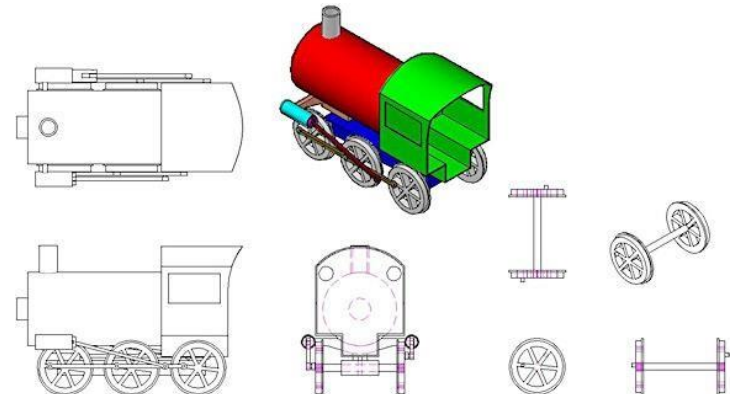
- Graphics programming packages provide functions to describe a scene in terms of basic geometric structures referred to as Output Primitives

- Output primitives include:**

- Points
- Lines
- Circles
- Spline curves



- Output primitives can be used to present more complex objects**





-3Ellipses Drawing Algorithms

- A- Scan Converting Ellipses
- **B- Midpoint Ellipse Algorithm**



Equation of an ellipse

- Consider an ellipse centered at the origin,
 $(x_c, y_c) = (0, 0)$:

$$\left(\frac{x}{a}\right)^2 + \left(\frac{y}{b}\right)^2 = 1$$

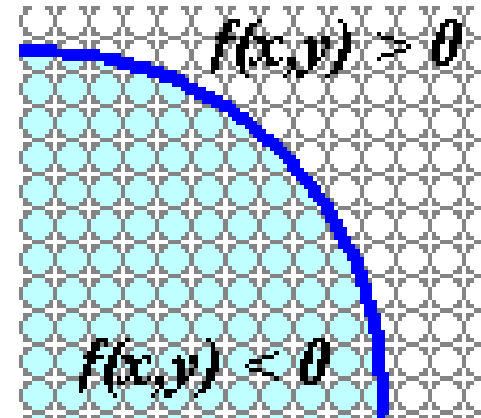
Ellipse function $f_e(x, y) = b^2 x^2 + a^2 y^2 - a^2 b^2$

...and its properties:

$f_{\text{ellipse}}(x, y) < 0$ if (x, y) is inside the ellipse

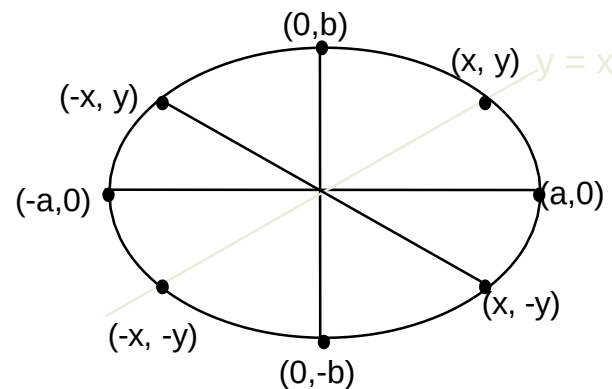
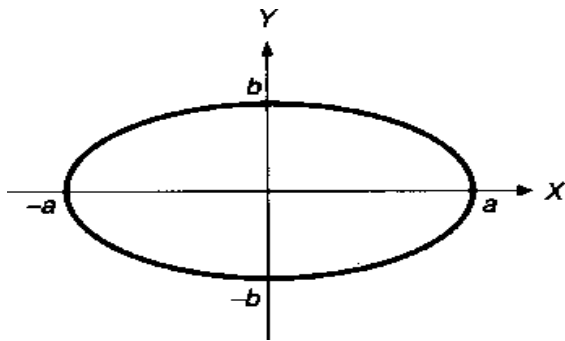
$f_{\text{ellipse}}(x, y) = 0$ if (x, y) is on the ellipse

$f_{\text{ellipse}}(x, y) > 0$ if (x, y) is outside the ellipse



A- Scan Converting Ellipses

$$F(x, y) = b^2 x^2 + a^2 y^2 - a^2 b^2 = 0$$



- $2a$ is the length of the major axis along the x axis.
- $2b$ is the length of the minor axis along the y axis.
- The midpoint can also be applied to ellipses.
- For simplicity, we draw only the arc of the ellipse that lies in the first quadrant, the other three quadrants can be drawn by symmetry

$$a = r_x, \quad b = r_y$$

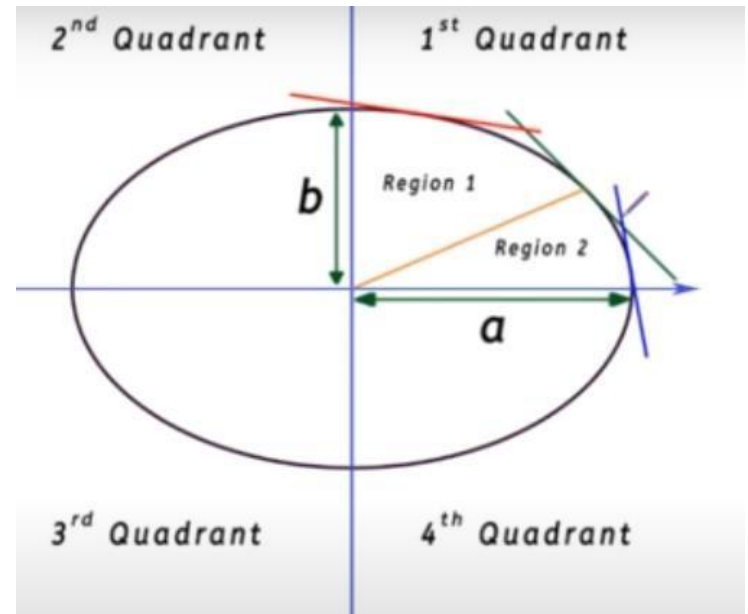
Properties of Ellipses

- Equation simplified if ellipse axis parallel to coordinate axis

$$\left(\frac{x - x_c}{a}\right)^2 + \left(\frac{y - y_c}{b}\right)^2 = 1$$

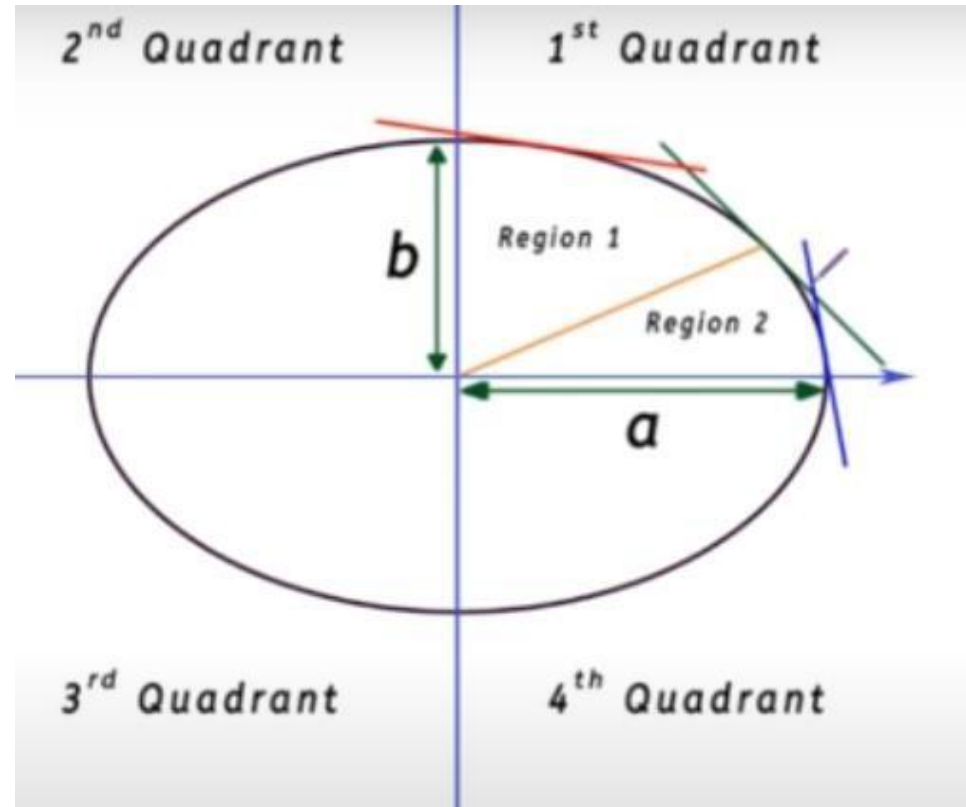
- Parametric form

$$\begin{aligned}x &= x_c + a \cos \theta \\y &= y_c + b \sin \theta\end{aligned}$$



B- Midpoint Ellipse Algorithm

- The center of the ellipse at $(0,0)$
- We will solve the algorithm at the first quadrant. **Points on the other quadrant will be computed from the first quadrant**
- **The first quadrant has two regions**
- If we draw a tangent at any point on the ellipse at region 1, the slope of the tangent must be less than 1 (**red line**)
- If we draw a tangent at any point on the ellipse at region 2, the slope of the tangent must be greater than 1 (**blue line**)



B- Midpoint Ellipse Algorithm

The green tangent is separating each region. The slope of this tangent is $m = -1$

The equation of an ellipse is $(x^2/a^2) + (y^2/b^2) = 1$

or, $x^2b^2 + y^2a^2 - a^2b^2 = 0 = f(x, y)$ ✓

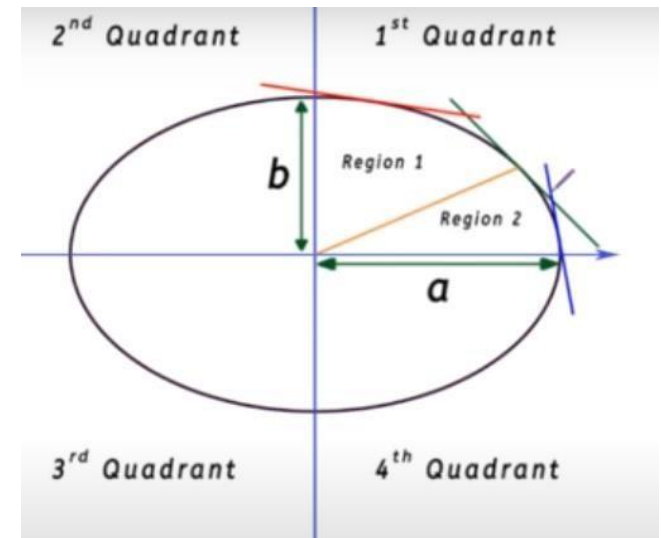
we know the slope of any line is $m = dy/dx$

the partial derivative of $f(x, y)$ w.r.t $x = f_x = 2xb^2$

the partial derivative of $f(x, y)$ w.r.t $y = f_y = 2ya^2$

$$dy / dx = - (f_x / f_y) = (- 2xb^2 / 2ya^2)$$

So when $2xb^2 \geq 2ya^2$ calculation for region 1 is stopped and for region 2 will start.



B- Midpoint Ellipse Algorithm

For the Region 1:

The slope of tangent ($m < 1$)

The x increases at unit interval so, $x_{k+1} = x_k + 1$

the y value will be either y_k or $y_k - 1$

So the next point will be either $(x_k + 1, y_k)$ or $(x_k + 1, y_k - 1)$

the midpoint = (x_m, y_m)

$$x_m = \{(x_k + 1) + (x_k + 1)\}/2 = x_k + 1$$

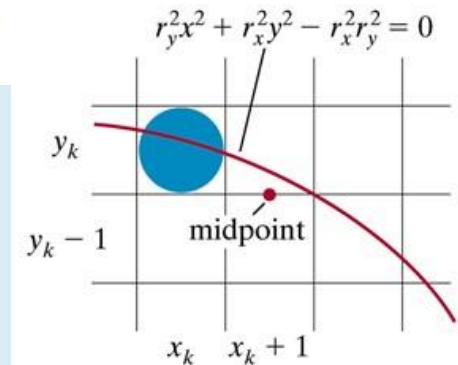
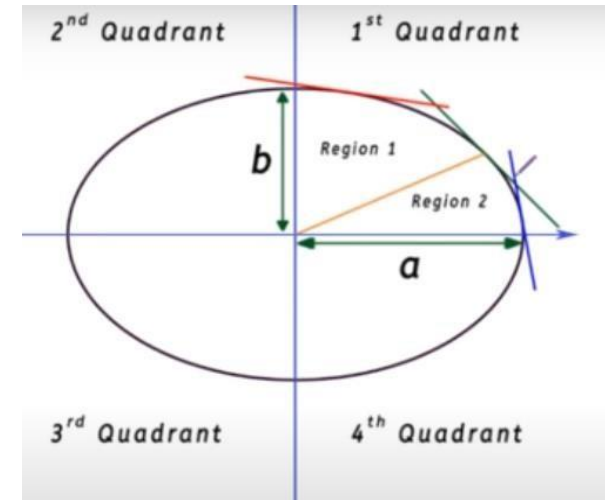
$$y_m = \{y_k + (y_k - 1)\}/2 = y_k - \frac{1}{2}$$

$$(x_m, y_m) = (x_k + 1, y_k - \frac{1}{2})$$

Putting the value of midpoint in the ellipse equation we get the decision parameter P_{1k}

$$P_{1k} = (x_k + 1)^2 b^2 + (y_k - \frac{1}{2})^2 a^2 - a^2 b^2$$

[Here P_{1k} is P_k for region 1]



Region 1:

$p_{1k} < 0$:

- midpoint is inside

• choose pixel (x_{k+1}, y_k) $p_{1k} \geq 0$:

- midpoint is outside/on
- choose pixel $(x_{k+1}, y_k - 1)$

B- Midpoint Ellipse Algorithm

$$P_{1k+1} = (x_{k+1} + 1)^2 b^2 + (y_{k+1} - \frac{1}{2})^2 a^2 - a^2 b^2$$

$$\text{or } P_{1k+1} = \{(x_k + 1) + 1\}^2 b^2 + (y_{k+1} - \frac{1}{2})^2 a^2 - a^2 b^2$$

now,

$$P_{1k+1} - P_{1k} = [\{(x_k + 1) + 1\}^2 b^2 + (y_{k+1} - \frac{1}{2})^2 a^2 - a^2 b^2] - [(x_k + 1)^2 b^2 + (y_k - \frac{1}{2})^2 a^2 - a^2 b^2]$$

$$\text{or } P_{1k+1} - P_{1k} = (x_k + 1)^2 b^2 + b^2 + 2(x_k + 1)b^2 + (y_{k+1})^2 a^2 + \frac{1}{4} a^2 - y_{k+1} a^2 - a^2 b^2 -$$

$$(x_k + 1)^2 b^2 - (y_k)^2 a^2 - \frac{1}{4} a^2 + y_k a^2 + a^2 b^2$$

$$\text{or } P_{1k+1} - P_{1k} = b^2 + 2(x_k + 1)b^2 + a^2\{(y_{k+1})^2 - (y_k)^2\} - a^2(y_{k+1} - y_k)$$

$$P_{1k+1} = P_{1k} + b^2 + 2(x_k + 1)b^2 + a^2\{(y_{k+1})^2 - (y_k)^2\} - a^2(y_{k+1} - y_k) \text{ [Decision parameter for first region]}$$

B- Midpoint Ellipse Algorithm

$$P_{1k} = (x_k + 1)^2 b^2 + (y_k - \frac{1}{2})^2 a^2 - a^2 b^2$$

The ellipse starts from (0, b), therefore putting (0, b) in P_{1k} we get,

$$P_{1k} = (0+1)^2 b^2 + (b - \frac{1}{2})^2 a^2 - a^2 b^2$$

$$\text{or } P_{1k} = b^2 + b^2 a^2 + \frac{1}{4} a^2 - a^2 b - a^2 b^2$$

or $P_{1k} = b^2 + \frac{1}{4} a^2 - a^2 b$ [Initial Decision parameter for first region]

Now, if $P_{1k} \geq 0$ then the next coordinate is $(x_k + 1, y_k - 1)$

else if $P_{1k} < 0$ then the next coordinate is $(x_k + 1, y_k)$

B- Midpoint Ellipse Algorithm

For the Region 2:

The slope of tangent ($m > 1$)

The y decreases at unit interval so, $y_{k+1} = y_k - 1$

the x value will be either x_k or $x_k + 1$

So the next point will be either $(x_k, y_k - 1)$ or $(x_k + 1, y_k - 1)$

the midpoint = (x_m, y_m)

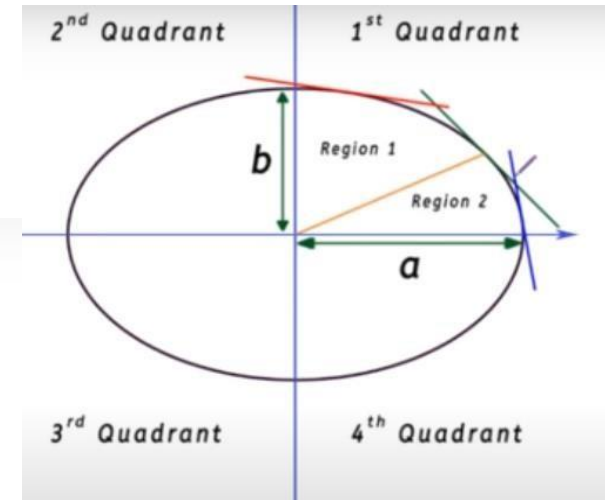
$$x_m = \{(x_k + (x_k + 1))\}/2 = x_k + \frac{1}{2}$$

$$y_m = \{(y_k - 1) + (y_k - 1)\}/2 = y_k - 1$$

$$(x_m, y_m) = (x_k + \frac{1}{2}, y_k - 1)$$

Putting the value of midpoint in the ellipse equation we get the decision parameter P_{2k}

$$P_{2k} = (x_k + \frac{1}{2})^2 b^2 + (y_k - 1)^2 a^2 - a^2 b^2 \quad [\text{Here } P_{2k} \text{ is } P_k \text{ for region 2}]$$



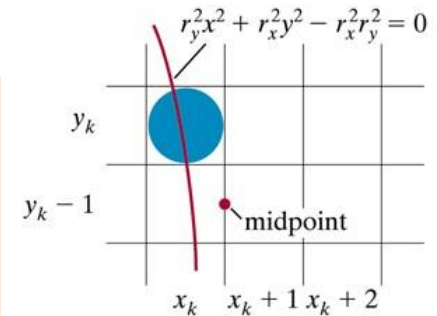
Region 2:

$p_{2k} \leq 0$:

- midpoint is inside/on
- choose pixel $(x_k + 1, y_k - 1)$

$p_{2k} > 0$:

- midpoint is outside
- choose pixel $(x_k, y_k - 1)$



B- Midpoint Ellipse Algorithm

$$P_{2k+1} = (x_{k+1} + \frac{1}{2})^2 b^2 + (y_{k+1} - 1)^2 a^2 - a^2 b^2$$

$$\text{or } P_{2k+1} = (x_{k+1} + \frac{1}{2})^2 b^2 + \{(y_k - 1) - 1\}^2 a^2 - a^2 b^2$$

now,

$$P_{2k+1} - P_{2k} = [(x_{k+1} + \frac{1}{2})^2 b^2 + \{(y_k - 1) - 1\}^2 a^2 - a^2 b^2] - [(x_k + \frac{1}{2})^2 b^2 + (y_k - 1)^2 a^2 - a^2 b^2]$$

$$\begin{aligned} \text{or } P_{2k+1} - P_{2k} &= (x_{k+1})^2 b^2 + \frac{1}{4} b^2 + x_{k+1} b^2 + a^2 (y_k - 1)^2 + a^2 - 2(y_k - 1)a^2 - a^2 b^2 \\ &\quad - (x_k)^2 b^2 - x_k b^2 - \frac{1}{4} b^2 - (y_k - 1)^2 a^2 + a^2 b^2 \end{aligned}$$

$$\text{or } P_{2k+1} - P_{2k} = a^2 - 2(y_k - 1)a^2 + b^2\{(x_{k+1})^2 - (x_k)^2\} + b^2(x_{k+1} - x_k)$$

$$P_{2k+1} = P_{2k} + a^2 - 2(y_k - 1)a^2 + b^2\{(x_{k+1})^2 - (x_k)^2\} + b^2(x_{k+1} - x_k) \text{ [Decision parameter for second region]}$$

The initial decision parameter value will be calculated after completing the first region.

Now, if $P_{2k} \geq 0$ then the next coordinate is $(x_k, y_k - 1)$

else if $P_{2k} < 0$ then the next coordinate is $(x_k + 1, y_k - 1)$



Midpoint Ellipse Algorithm

- (1) Input r_x , r_y and ellipse centre (x_c, y_c) obtain the first point $(x_0, y_0) = (0, r_y)$

$$a = r_x, \quad b = r_y$$

- (2) Calculate the initial value of the decision parameter in region 1 as

$$P_{10} = r_y^2 - r_x^2 r_y + \frac{1}{4} r_x^2$$

- (3) At each x_k position in region 1, starting at $K = 0$. If $P_k < 0$, then next point along ellipse is (x_{k+1}, y_k) and $P_{k+1} = P_k + 2r_y^2 x_{k+1} + r_y^2$ otherwise next point along the circle is $(x_k + 1, y_k - 1)$ and $P_{k+1} = P_k + 2r_y^2 x_{k+1} - 2r_x^2 y_{k+1} + r_y^2$ with $2r_y^2 x_{k+1} = 2r_y^2 x_k + 2r_y^2$, $2r_x^2 y_{k+1} = 2r_x^2 y_k - 2r_x^2$ and continue until $2r_y^2 x \geq 2r_x^2 y$.

- (4) Calculate the initial value of the decision parameter in region (2) using last point (x_0, y_0) calculated in region 1 as $P_{20} = r_y^2 (x_0 + \frac{1}{2})^2 + r_x^2 (y_0 - 1)^2 - r_x^2 r_y^2$.

- (5) At each y_k position in region (2), starting at $k = 0$ perform following test : If $P_{2k} > 0$ next point is $(x_k, y_k - 1)$ and $P_{2k+1} = P_{2k} - 2r_x^2 y_{k+1} + r_x^2$ otherwise next point is $(x_k + 1, y_k - 1)$ and $P_{2k+1} = P_{2k} + 2r_y^2 x_{k+1} - 2r_x^2 y_{k+1} + r_x^2$.

- (6) Determine the symmetry points in other three quadrants.

- (7) More each calculated pixel position (x, y) . Center on (x_c, y_c) , plot coordinate values $x = x + x_c$ & $y = y + y_c$.

- (8) Repeat the steps for region 1 until $2r_y^2 x \geq 2r_x^2 y$.



– Example 1

– Given $r_x = 8$ and $r_y = 6$, $r_x^2 = 64$ and $r_{y_2} = 36$

$$P_{10} = f_{\text{ellipse}}(x_0, y_0) = (1, r_y - 1/2) = r_y^2 + r_x^2 (r_y - 1/2)^2 - r_x^2 r_y^2$$

$$P_{10} = r_y^2 - r_x^2 r_y + 1/4 r_x^2$$

At each x_k position in region 1, starting at $K = 0$. If $P_k < 0$, then next point along ellipse is (x_{k+1}, y_k) and $P_{k+1} = P_k + 2r_y^2 x_{k+1} + r_y^2$ otherwise next point along the circle is $(x_k + 1, y_k - 1)$ and $P_{k+1} = P_k + 2r_y^2 x_{k+1} - 2r_x^2 y_{k+1} + r_y^2$ with $2r_y^2 x_{k+1} = 2r_y^2 x_k + 2r_y^2$, $2r_x^2 y_{k+1} = 2r_x^2 y_k - 2r_x^2$ and continue until $2r_y^2 x \geq 2r_x^2 y$.

First point in region 1 is (0, 6)

$$r_x^2 = 64, \quad r_y^2 = 36$$

$$2r_x^2 = 128, \quad 2r_y^2 = 72$$

$$P_{10} = 36 - (64 \times 6) + \frac{1}{4} \times 64 = 36 - 384 + 16 = -332$$

$p < 0$

So, next point (increment x and keep y)

$P_{10} < 0 \therefore$ Next point is (1, 6)

$$P_{11} = -332 + 72 \times 1 + 36 = -332 + 108 = -224$$

$p < 0$

So, next point (increment x and keep y)

$P_{11} < 0 \therefore$ Next point is (2, 6)



– Example 1

– Given $r_x = 8$ and $r_y = 6$, $r_x^2 = 64$ and $r_y^2 = 36$

$$P_{10} = f_{\text{ellipse}}(x_0, y_0) = (1, r_y - 1/2) = r_y^2 + r_x^2(r_y - 1/2)^2 - r_x^2 r_y^2$$

$$P_{10} = r_y^2 - r_x^2 r_y + 1/4 r_x^2$$

At each x_k position in region 1, starting at $K = 0$. If $P_k < 0$, then next point along ellipse is (x_{k+1}, y_k) and $P_{k+1} = P_k + 2r_y^2 x_{k+1} + r_y^2$ otherwise next point along the circle is $(x_k + 1, y_k - 1)$ and $P_{k+1} = P_k + 2r_y^2 x_{k+1} - 2r_x^2 y_{k+1} + r_y^2$ with $2r_y^2 x_{k+1} = 2r_y^2 x_k + 2r_y^2$, $2r_x^2 y_{k+1} = 2r_x^2 y_k - 2r_x^2$ and continue until $2r_y^2 x \geq 2r_x^2 y$.

$p < 0$

So, next point (increment x and keep y)

$$P_{12} = -224 + 72 \times 2 + 36 = -224 + 144 + 36 = -44$$

$P_{12} < 0 \therefore$ Next point is (3, 6)

$$P_{13} = -44 + 72 \times 3 + 36 = -44 + 216 + 36 = 208 \quad \text{p>0 So, next point (increment x and decrement y)}$$

$P_{13} > 0 \therefore$ Next point is (4, 5)

$$P_{14} = 208 + 72 \times 4 - 128 \times 5 + 36 = 208 + 288 - 640 + 36 = -108 \quad \text{p<0 So, next point (increment x and keep y)}$$

$P_{14} < 0 \therefore$ Next point is (5, 5)

$$P_{15} = -108 + 72 \times 5 + 36 = 288$$

$p > 0$ So, next point (increment x and decrement y)

$P_{15} > 0 \therefore$ Next point is (6, 4)

$$P_{16} = 288 + 72 \times 6 - 128 \times 4 + 36 = 244$$

$p > 0$ So, next point (increment x and decrement y)

$P_{16} > 0 \therefore$ Next point is (7, 3)

$$P_{17} = 244 + 72 \times 7 - 128 \times 3 + 36 = 1168$$



Example 1
 Given $r_x = 8$ and $r_y = 6$, $r_x^2 = 64$ and $r_y^2 = 36$

$P1_k$	(x_{k+1}, y_{k+1})	$2r_y^2 x_{k+1}$	$2r_x^2 y_{k+1}$
- 332	(1, 6)	72	768
- 224	(2, 6)	144	768
- 44	(3, 6)	216	768
208	(4, 5)	288	640
- 108	(5, 5)	360	640
288	(6, 4)	432	512
244	(7, 3)	504	384

Stop region 1 at (7, 3)
 because $504 > 384$

$$P2_0 = f_{\text{ellipse}}(x_0 + 1/2, y_0 - 1) = r_y^2(x_0 + 1/2)^2 + r_x^2(y_0 - 1)^2 - r_x^2 r_y^2$$

Now we move to region 2 since $2r_x^2 > 2r_x^2 y$

First point in region 1 is (0, 6)

Now For region the initial point is $(x_0, y_0) = (7, 3)$ the initial decision parameter is

$$P2_0 = f(7 + 1/2, 3 - 1) = 36(7 + 1/2)^2 + 64(2)^2 - 64 \times 36 = 2025 + 256 - 2304 = -23$$

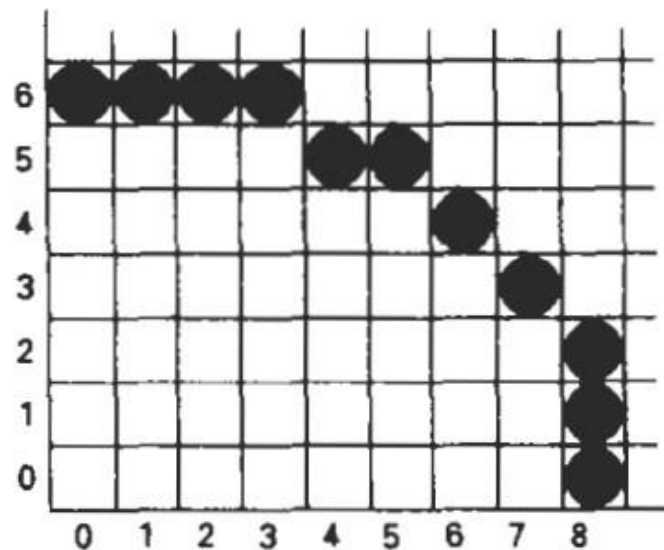


Example 1
 Given $r_x = 8$ and $r_y = 6$, $r_x^2 = 64$ and $r_y^2 = 36$

The remaining positions are then calculated as :

K	P1 _k	(x _{k+1} , y _{k+1})	2r _y ² x _{k+1}	2r _x ² y _{k+1}
0	- 23	(8, 2)	576	256
1	- 215	(8, 1)	576	128
2	- 279	(8, 0)	-	-

Now Plot these point on the graph.



Example 2

Given an ellipse with $r_x = 2$ and $r_y = 4$ and center (4, 5), trace the mid-point ellipse drawing algorithm for points.

Hints:

$$p1_0 = r_y^2 - r_x^2 r_y + \frac{1}{4} r_x^2$$

$$\text{increment} = \begin{cases} 2r_y^2 x_{k+1} + r_y^2 & \text{if } p1_k < 0 \\ 2r_y^2 x_{k+1} + r_y^2 - 2r_x^2 y_{k+1} & \text{if } p1_k \geq 0 \end{cases}$$

k	x_k	y_k	p_k
0	0	4	1
1	1	3	-15
2	2	3	33
3	3	2	89
4	4	1	

$$\begin{aligned} r_y^2 &= 16 \\ r_x^2 &= 4 \\ r_x^2 r_y &= 16 \end{aligned}$$

Initial point : $(0, r_y) = (0, 4)$; $p > 0$
So, next point (increment x and decrement y)

$$\begin{aligned} p1_0 &= r_y^2 - r_x^2 r_y + 1/4 r_x^2 = 16 - 16 + 4/4 = 1 \\ p1_1 &= p_0 + 2r_y^2 x_{k+1} + r_y^2 - 2r_x^2 y_{k+1} = 1 + (2*16*0) + 16 - (2*4*4) = -15 \\ p1_2 &= p_1 + 2r_y^2 x_{k+1} + r_y^2 = -15 + (2*16*1) + 16 = 33 \\ p1_3 &= p_2 + 2r_y^2 x_{k+1} + r_y^2 - 2r_x^2 y_{k+1} = 33 + (2*16*2) + 16 - (2*4*3) = 89 \\ p1_4 &= p_3 + 2r_y^2 x_{k+1} + r_y^2 - 2r_x^2 y_{k+1} = 89 + (2*16*3) + 16 - (2*4*2) = 185 \end{aligned}$$

$p < 0$
So, next point (increment x and keep y)